

got so high that a regular commercial building just couldn't handle it. There are innumerable examples of CIOs and CTOs who have talked of this problem — about not being able to have that kind of power in their data centre. They have cooling problems they cannot handle because the businesses are scaling up, big time. Even if they want to put more racks to accommodate more servers, they don't have the space, with offices all around. These are the challenges that CIOs and CXOs are facing. And that's why there is so much of migration among customers — they are closing their captive data centres and moving to third party data centres, which are giving them the same power, cooling and scalability, all in a much better way.

The second big challenge is that the businesses are growing crazily. Companies in e-commerce are growing really fast. They place requirements in front of the IT team to launch an application practically the next day. There is also a budget constraint put in by the CFO of the company. So one ultimately ends up 'oversizing the hardware' and this is the typical standard problem for all CIOs and CTOs. The solution to this is the cloud, and this is why this big wave of cloud adoption has happened.

You can just go and purchase a cloud instance of a reasonable size, and launch the application tomorrow without bothering about sizing beyond a point, because if you find that the cloud purchased is not meeting your requirement, you can actually increase that at a click of a button. And if you think you have taken a bigger infrastructure than needed, you can actually reduce it with the click of a button. The best part is that you are not stuck with a particular technology.

Q What are the kind of questions an IT head has to ask to cloud service providers while investing in the cloud?

Essentially, the first question should be, 'Can I adopt the cloud?' Different types of customers may have different needs. There are customers who are savvy, having dabbled with this technology. Such customers do not take time to adopt it, as the cloud is also becoming a commodity now; there are so many customers in the e-commerce domain who have become so very comfortable with the cloud that they don't ask any questions, they just ask us to deliver. However, at the other end of the spectrum, there are still people in traditional enterprises like those in the manufacturing domain, who have a lot of questions. The questions do not really start with a transactional analysis of the cloud offering. They include, "Can you advise me on whether I can move to the cloud?"; "If yes, what applications can I move to the cloud?"; "If not, why can't I move to the cloud?" or "How exactly can I migrate to the cloud?" They want to see the entire way forward.

The questioning from such clients is very basic, and these are the right questions to ask before getting into the nitty gritty of the cloud.

Q What are the kind of workloads that are suitable for the cloud?

Frankly speaking, if this question had been asked five years back, my answer would have been different. But today, I would say that cloud technologies have evolved and matured so much that any type of critical workload that demands huge resources in terms of compute, storage and networks can be migrated to the cloud. In general, anything which runs on Windows or any flavour of Linux can easily be migrated to the cloud.

Q So what would you like to tell clients who are looking for cloud computing to improve the utilisation of their existing servers, but are also concerned whether the utilisation and optimisation will compromise the service level agreements?

First of all, both the technology and the cloud service provider model, as they exist today, are so very evolved. When the technology came in six or seven years back, this would have been a good topic of debate. But nowadays, when clients invest in a cloud grid (which may host multiple customers), they get far more redundancy, flexibility and scalability than they might want. The amount of redundancy, security, flexibility and scalability available with a cloud grid today is far higher compared to the standard one-on-one solution for a customer. So the service levels that are given on a cloud grid nowadays are at par and even better than what was offered on a simple dedicated hardware type of solution earlier.

Q What is the importance of multi-tenancy in the cloud paradigm?

The entire underlying basis of the cloud is multi-tenancy. Every single enterprise has hundreds of applications, for which it could end up investing in hundreds of servers, storage boxes, network boxes, et al. According to several research studies, not more than 30-35 per cent of the capacity, which is invested in, is ever utilised. And the balance 70 per cent capacity is just a waste of money. And that waste of money is not only in hardware, but also an investment in costly software licences. Besides, accordingly, you end up investing so much money in creating data centres, power, cooling, manpower, networking...and a whole lot more on everything. It's a huge waste of effort too.

What the cloud ultimately did was, instead of thousands of enterprises creating their own individual environments where there is so much wastage of resources, it created a centralised grid with a pool of resources. It cuts these resources into 'pieces' and gives it to individual customers. So when this is happening, if say, one customer is not using some resources, these are available for other customers to use. And that is where you are bringing in the optimisation. So the basic premise of the cloud is based on multi-tenancy, where a centralised service provider makes a huge investment in creating a vast pool of resources like compute, storage or networks. 